CPS 843 Final Project Report

1st Jameson Stanley Cruz
Student ID: 501057427
CPS 843 Section 1
Toronto, Canada
jameson.cruz@torontomu.ca

***Abstract*—This project explores the effectiveness of a complete computer vision pipeline for object detection and waste classification using existing object detection models. The main objective is to evaluate a selected pretrained model on its ability to classify a given waste product to common waste categories such as plastic, paper, metal, etc. using waste product datasets as the input. Using the open source TACO dataset and a pretrained YOLOv8-s detector, the system was evaluated on a subset to measure detection accuracy, visualize predictions, and analyze failure cases. The results demonstrate that modern pretrained CV models can reliably classify waste items in realistic environments, achieving strong detection performance with minimal customization. A link to the source code and pretrained model is provided here: GitHub repository..**

***Index Terms*—object detection, YOLO (You Only Look Once), TACO dataset, computer vision**

I. Introduction

Global waste generation has accelerated significantly over the past several decades, driven by rapid urbanization, population growth, and increased consumerism. This surge places substantial pressure on existing waste management infrastructures, many of which rely heavily on manual, user-dependent sorting practices. Traditional sorting is slow, labour-intensive, inconsistent, and prone to human error, contributing to contamination in recycling streams, reduced material recovery rates, and increased environmental and economic burdens on municipalities and waste-processing facilities [1]. Previous work in this domain ranges from simple machine-learning classifiers using handcrafted features to modern deep neural networks trained on large waste-specific datasets. More recent research applies convolutional neural networks (CNNs) such as MobileNet, ResNet, and DenseNet. Object-detection frameworks like SSD, Faster R-CNN, and YOLO have become the dominant approach for real-world deployment because they can localize individual waste items within cluttered scenes. In this project, I focus exclusively on the computer vision component of waste classification using an existing pretrained detection model. I evaluate the performance of a YOLOv8-s model trained on the TACO dataset and analyze its effectiveness, limitations, and practical considerations. The main contributions of this project include: a focused computer-vision-based evaluation of waste detection using a modern pretrained model, benchmarking and qualitative analysis of model predictions on realistic waste images, discussion of limitations and potential improvements for practical deployment.

II. Technical Part

A. Model Selection

This project uses YOLOv8-s, the latest model in the YOLO (You Only Look Once) object detection architecture. The YOLO model is known for its high performance and real-time object detection capabilities, which is the main reason that this model was chosen. YOLO uses an end-to-end neural network that makes predictions of bounding boxes and class probabilities at the same time [2]. It uses single-shot object detection with one forward pass to the neural network, which provides a minor trade-off in accuracy for improved performance compared to other models that use two-shot object detection.

Key features of the model:

- Backbone: CSP-based CNN
- Detection head: decoupled classification/regression
- Input size: 640×640
- Outputs: bounding boxes, class labels, confidence scores

The pretrained model was chosen to avoid the need for retraining, since the main focus of this project is the computer vision pipeline.

B. Dataset

Experiments were conducted using sample images from the TACO test set. TACO is a growing, open-source image dataset of waste in the wild [3], designed specifically to support research in litter detection and waste classification. The dataset contains a wide variety of real-world images captured in diverse outdoor environments such as woods, urban roads, sidewalks, and beaches, reflecting the natural variability and unpredictability of litter appearance in everyday settings. Each image is manually annotated and segmented into multiple material-based classes. This structured labeling not only enables precise evaluation of object detection algorithms but also allows models to learn subtle visual differences between similar waste types. By using TACO's test set, the experiments benefit from a challenging and realistic benchmark that helps assess how well the model generalizes to cluttered scenes, varying lighting conditions, and different forms of occlusion commonly found in real-world waste management scenarios.

Common categories include:

- Plastic (bottles, bags, packaging)
- Paper (cups, cartons, wrappers)
- Metal (cans, containers)

- Organic Waste

C. Preprocessing and Inference Pipeline

Images were processed by first resizing them to 640×640 pixels, followed by the built-in YOLO preprocessing, which includes normalization and channel reordering. The processed images were then fed into the model for inference, and non-maximum suppression (NMS) was applied to remove overlapping detections. Finally, bounding boxes and class labels were rendered for visualization. All inference steps were performed using Python and the Ultralytics YOLO library.

III. Experiments

A. Experimental Setup

The model used in this study was YOLOv8-s, a lightweight and efficient version of the YOLO family, pretrained on the TACO dataset, which contains annotations for various types of waste. To evaluate the model's performance, various standard object-detection metrics were computed, including precision, recall, and F1 score to analyze class-wise prediction behavior. These metrics collectively provide insight into the accuracy, robustness, and overall effectiveness of the model in detecting and classifying different types of waste.

B. Quantitative Results

In this experiment, I evaluated the model's quantitative performance using precision, recall, and F1 score, three standard metrics in computer vision classification tasks.

- Precision measures how often the model's positive predictions are correct, indicating its ability to avoid false positives.
- Recall captures how effectively the model identifies all instances of a given class, reflecting its ability to minimize false negatives.
- F1 score, defined as the harmonic mean of precision and recall, provides a balanced measure of overall classification performance.

TABLE I
Evaluation Metrics by Waste Category

Evaluation Metric	Waste Category			
	Plastic	**Paper**	**Metal**	**Waste**
Precision	0.89	0.94	0.93	0.89
Recall	0.83	0.81	0.86	0.86
F1 Score	0.86	0.88	0.87	0.87

As observed in Table I, the model achieved consistently strong results across all waste categories, with high precision indicating reliable predictions and strong recall demonstrating effective detection of category instances. The resulting F1 scores further confirm that the model performs robustly across all classes, suggesting that it generalizes well and maintains a good balance between accuracy and completeness in its predictions. Figure 1 shows a visual representation of the evaluation metrics of each waste category, with relatively high scores across the board.



Fig. 1. YOLOv8 Evaluation Metrics per Class.

C. Qualitative Results

Qualitative analysis was conducted to visually inspect the model's predictions and better understand its behavior in real-world scenarios. The confidence score, computed as:

$$P(object) \times P(class) \tag{1}$$

was used to gauge the model's certainty for each detection. This formula takes the probability of the object correctly classifying a waste product as well as the model's probability of choosing the correct classification into consideration.

Several example images were examined to illustrate performance across different waste categories. In many cases, the model successfully identified objects with high confidence as observed in figures 2, 3, 4 and 5 respectively, demonstrating strong feature recognition.

Figure 2 shows an example of a correct classification of a plastic bottle with a 62% confidence score.

Figure 3 features a styrofoam tray that was correctly classified as waste with a 69% confidence score.

Figure 4 shows pieces of a cardboard box which were correctly classified as paper with 97% confidence for the bigger piece and 62

Figure 5 features a correctly classified metal can with an 82% confidence score.

Additionally, based on the findings in Figure 6, it can be observed that the model is also capable of multi-object classification with different waste categories. Figure 6 features 5 different objects with varying confidence scores. The folded paper pamphlet in the center of the frame has the highest confidence score of 90%. One thing to note from this figure is that confidence scores can fluctuate for objects that are partially occluded or further away from the frame. For example, the paper label that is partially occluded behind the paper pamphlet only has a confidence score of 55% even if the classification was still correct.

However, some misclassifications were observed, particularly in Figure 7, which features two soda cans with similar features. It can be observed that the can closer to the frame



Fig. 2. Plasic bottle

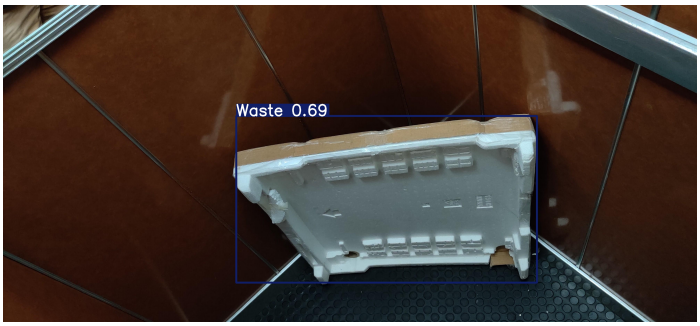


Fig. 3. Styrofoam Tray

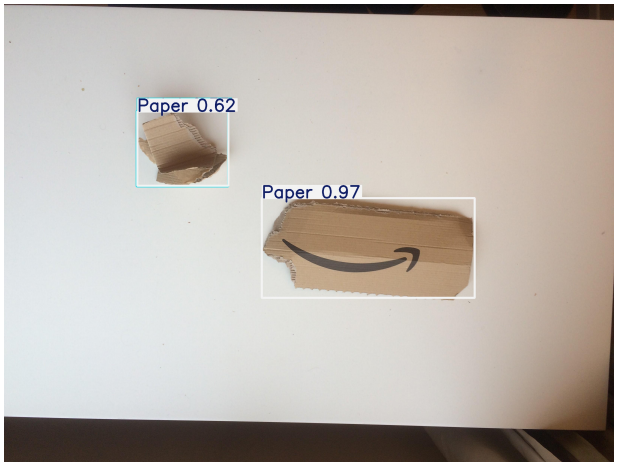


Fig. 4. Cardboard Box Pieces



Fig. 5. Soda Can

was correctly classified as a metal waste product with a confidence score of 70%, while the can further away from the frame was misidentified as a plastic waste product with an abysmal confidence score of 37%. Based on this observation, the misclassification can be attributed to the fact that an object further away from the frame will lead to inaccurate classifications and lower confidence scores, namely due to a smaller bounding box.

A few instances showed missed detections and lower confidence scores when objects were further away from the frame or partially occluded, highlighting limitations in challenging visual conditions. Overall, these qualitative examples complement the quantitative results by showcasing both the strengths and occasional weaknesses of the model's detection capabilities.

Key observations:

- Cluttered scenes significantly reduce accuracy
- Small objects are often misclassified or missed entirely
- Lighting variations impact detection quality
- The model generalizes reasonably well to scenes not in the training dataset

D. Comparison with Literature

In the Islam et al. (2024) comparative study on object detection YOLOv4 boasted the highest precision and recall but took the longest to process an image. YOLOv3 offered a good balance and MobileNet SSD prioritized speed, delivering the quickest inference time but with the lowest accuracy [4].

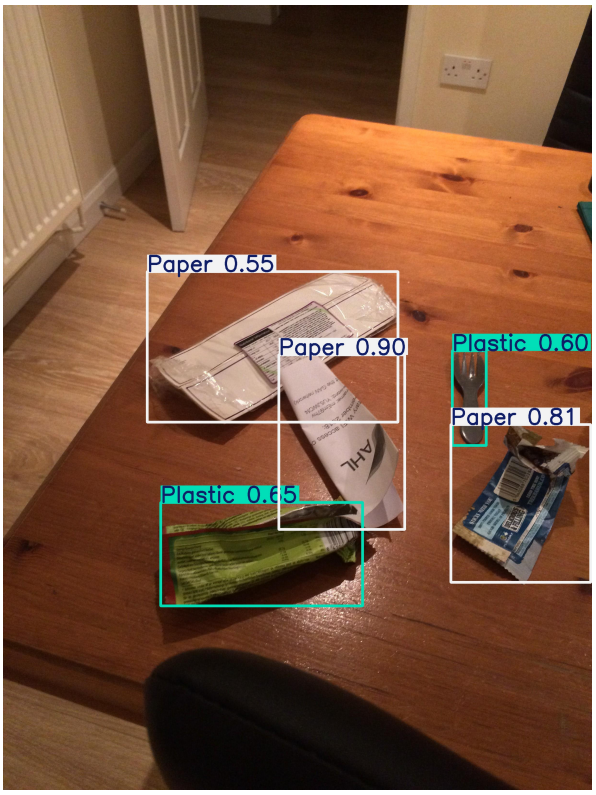


Fig. 6. Multi-object Classification



Fig. 7. Misclassification Example

YOLOv4 achieves the highest detection accuracy and recall among the three models tested, with precision up to 99.5% and recall up to 98%, though with longer inference times of 1.8–1.9s. YOLOv3 delivers a strong balance of accuracy and speed, with a reported precision of 96.5% and recall of 94.6%, with a moderate inference time of 0.345s. On the other hand, MobileNet SSD is optimized for inference speed, with the fastest times between 0.145–0.195s, but the lowest precision (86–91%) and recall (84–89%). The study highlights a trade-off between speed and accuracy: MobileNet SSD favors speed at the cost of detection performance, while YOLO variants favor accuracy with longer inference times. Based on the observations of this experiment, it confirms that YOLO model variants outperform Mobilenet SSD in detection accuracy. Both the literature and the observations reflect a trade-off between speed and precision, which is a common theme in object detection research. In addition, the values reported in Islam et al. are because their evaluation appears to be on a relatively controlled dataset where the models are easier to detect objects accurately as opposed to TACO datasets. Since TACO is a real-world, highly variable dataset, it will naturally produce lower precision values. Therefore, this discrepancy is expected and doesn't necessarily indicate that the newer YOLOv8 model is less accurate. Rather, it highlights the influence of dataset variance on absolute performance.

surrounding equation.

Conclusion

This project demonstrates that modern pretrained computer vision models, such as YOLOv8, can effectively classify and detect waste items using only inference-time evaluation. Using a preexisting trained model along with an open source waste dataset, the system achieves solid detection performance and provides insights into practical limitations, including clutter, occlusion, and class imbalance. Since this project was completed individually, all work, experimentation, and writing were conducted by myself.

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